

A Single Effective Framework Underlying Galaxy Rotation Curves and the Hubble Tension

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Abstract

We investigate whether two long-standing observational anomalies—flat galaxy rotation curves and the Hubble constant tension—may originate from a common effective mechanism operating in low-density environments. We introduce a minimal phenomenological framework in which departures from Newtonian expectations arise from a bounded saturation of the effective gravitational response at the level of kinematic inference, without invoking dark matter particles or modifying the background Friedmann expansion or early-universe physics. In the galactic regime, the MOND phenomenology emerges as the low-acceleration limit of this bounded-response framework rather than as a fundamental modification of gravity. The description reduces to standard Newtonian behavior in high-density environments and introduces a smooth, finite effective response in diffuse systems. Applied to disk galaxies, the framework reproduces the main features of observed rotation curves across a broad range of morphologies using baryonic matter alone, with stable parameter values and no halo-by-halo tuning. Explicit comparisons with representative galaxies from the SPARC sample show that the effective saturation scale correlates with observed surface-density-dependent trends. At cosmological scales, the same bounded-response structure predicts environment-dependent shifts in the locally inferred expansion rate. For typical void density contrasts, an explicit order-of-magnitude estimate demonstrates that the deep-saturation regime applies, yielding a fractional shift $\Delta H/H \simeq \delta_\rho f_{\max}$ without introducing additional cosmological degrees of freedom or altering the global expansion history. Cosmic voids thus provide a natural setting for a systematic offset between local and global determinations of the Hubble constant. We examine robustness, degeneracies with standard astrophysical effects, and limitations of the effective description, and outline distinctive observational signatures, including environment-dependent redshift drift and subtle weak-lensing modifications in extreme low-acceleration regimes. These results suggest that galaxy rotation curves and the Hubble tension may reflect a shared low-density phenomenology testable with current and forthcoming observations.

Keywords: cosmology; galaxy rotation curves; Hubble tension; modified gravity; voids

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1. Introduction

Two observational puzzles continue to motivate scrutiny of late-time gravitational phenomenology.

The first is the ubiquity of flat galaxy rotation curves. Across a wide range of disk galaxies, circular velocities remain nearly constant at large radii, in apparent tension with the expectation from Newtonian dynamics applied to the observed baryonic mass distribution. This behavior has been firmly established by large and homogeneous datasets,

most notably the SPARC compilation, which provides resolved rotation curves together with self-consistent baryonic mass models for disk galaxies [1]. Within the standard cosmological framework, flat rotation curves are typically accounted for by extended dark matter halos. While this approach is empirically successful, it introduces substantial model freedom at the level of individual galaxies and leaves open the question of why tight empirical regularities, such as surface-density-dependent trends and acceleration-based scalings, emerge so prominently in the data [2]. Historically, these regularities led to the development of the Modified Newtonian Dynamics (MOND) paradigm [3,4], which remains the primary phenomenological benchmark for such anomalies.

The second puzzle is the Hubble constant tension. Late-time local determinations of H_0 based on distance ladder techniques and standard candles remain in significant disagreement with early-time inferences derived from the homogeneous high-redshift Universe as constrained by cosmic microwave background observations [5,6]. This discrepancy has persisted across multiple datasets and analysis methods, prompting proposals ranging from unidentified systematics to extensions of the standard cosmological model. Comprehensive reviews indicate that no consensus explanation has yet emerged [7]. A notable feature of the tension is its strong association with late-time inference in a structured and inhomogeneous Universe.

Although galaxy rotation curves and the Hubble tension are usually discussed separately, both phenomena probe kinematic inference in low-density regimes. Galaxy outskirts and ultra-diffuse systems explore weak-acceleration regions, while cosmic voids represent the most extreme low-density environments on cosmological scales. This commonality motivates the central question addressed in this work. Can a single effective mechanism, active primarily in diffuse environments, provide a consistent phenomenological interpretation of both classes of anomalies within a unified descriptive framework.

In this paper, we adopt a perspective in which late-time cosmological observables are interpreted as effective quantities arising from a coarse-grained description of admissible macroscopic states. Within this view, quantities such as gravitational acceleration or the Hubble parameter are not treated as fundamental dynamical variables. They characterize effective transition rates between macrostates of the observable Universe under a given resolution of description.

Departures from Newtonian expectations in low-density environments are then not attributed to a modification of the force law, but to a limitation of the effective kinematic inference applied to increasingly diffuse configurations. They reflect a saturation of the effective description when further refinement of the inferred response becomes impossible. In this regime, the effective gravitational response does not continue to grow indefinitely as density decreases, but instead approaches a plateau controlled by an intrinsic saturation scale.

While the mathematical structure of this saturation in the galactic limit shares similarities with the a -field theories of the Bekenstein–Milgrom formulation [4], the present approach is conceptually distinct and does not posit a modification of the underlying equations of motion. The saturation does not represent an ad hoc modification of gravity. It arises from a limitation of the effective description applicable in diffuse regimes, where kinematic inference becomes progressively insensitive to further reductions in density.

This phenomenological study is motivated by a broader theoretical context in which bounded-response regimes of the Born–Infeld type arise naturally in effective descriptions of gravitational dynamics [8]. Related structural and projection-based considerations are discussed in companion works focusing on the informational foundations of the geometric support [9,10]. Here we deliberately restrict attention to late-time observational consequences and do not rely on the full underlying framework.

The model is constructed to reduce to standard Newtonian behavior in high-density environments and to introduce a smooth bounded modification in diffuse regimes. It is treated explicitly as an effective description of late-time phenomenology. No modification of early-universe physics is assumed, and the model is not presented as a fundamental theory of gravity.

The analysis proceeds in two stages. First, we test the model against galaxy rotation curve data, focusing on the SPARC sample, and evaluate whether a single universal saturation scale can reproduce the observed diversity of rotation curve shapes using baryonic matter alone, without halo-by-halo tuning. Second, we explore the cosmological implications of the same mechanism in low-density environments, emphasizing cosmic voids as maximal probes and deriving testable signatures for environment-dependent expansion inference.

A key aim of this work is falsifiability. Beyond fitting existing data, we identify observational discriminants that can confirm or exclude the framework. These include correlations between locally inferred values of H_0 and large-scale environment, redshift-dependent suppression of the local offset, void-specific redshift drift, and weak lensing signatures.

The structure of the paper is as follows. Section 2 introduces the effective model and its minimal assumptions. Section 3 presents rotation curve tests and diagnostics. Section 4 discusses the implications for cosmic voids and the Hubble tension. Section 5 examines robustness, degeneracies, and limits. Section 6 summarizes predictions and observational tests. Section 7 concludes.

2. Effective Model

This section introduces a minimal effective framework designed to capture late-time gravitational phenomenology in low-density environments. The model is constructed to address galaxy rotation curves and the Hubble tension within a single unified description, without modifying early-universe physics or invoking dark matter particles in this effective regime.

The emphasis is deliberately phenomenological. Only the minimal assumptions required to reproduce the observed anomalies are introduced. The model is not presented as a fundamental theory of gravity, but as an effective description valid in specific environmental regimes.

The guiding interpretation adopted here is that late-time gravitational observables characterize effective transition rates between admissible macroscopic configurations of the observable Universe. In this view, quantities such as gravitational acceleration or the Hubble parameter do not represent fundamental dynamical fields. They encode how rapidly the effective state of the Universe can be updated under a finite resolution of the underlying descriptive structure.

Saturation is interpreted as a limitation of this effective resolution. In low-density environments, the number of independent relations contributing to the gravitational response becomes bounded, so that further decreases in density no longer translate into proportionally stronger inferred dynamics. As a result, the effective response approaches a plateau rather than vanishing asymptotically. The dynamical motivation for bounded-response regimes of this type is developed in a companion theoretical work [8].

2.1. Minimal Assumptions and Domain of Validity

We assume that gravitational phenomenology at galactic and sub-cosmological scales can be described by an effective potential Φ_{eff} , used here as a convenient parametrization of

kinematic inference rather than as a fundamental dynamical field. The framework satisfies the following requirements.

(i) The effective dynamics must reduce to standard Newtonian gravity in high-density environments, ensuring consistency with Solar System tests. (ii) Deviations are allowed only in diffuse regimes, characterized by low baryonic density or weak gravitational gradients. (iii) The transition is governed by at most one additional universal scale parameter a_* , with no environment-specific tuning. (iv) Locality and isotropy are preserved at the effective level.

Mathematically, the framework shares the Lagrangian structure of the quasi-linear MOND theories proposed by Bekenstein and Milgrom [4]. However, whereas MOND introduces this modification as a fundamental alteration of the gravitational law, the present approach interprets it as a saturated effective response. The modification reflects a finite capacity of the effective description to resolve increasingly dilute relational structure [10].

This conceptual shift is essential. It allows the phenomenological success of MOND at galactic scales to be reinterpreted as a regime of descriptive saturation, rather than as a modification of gravity. The same mechanism then applies naturally to volumetric density dilution in cosmological voids.

The present framework is therefore not intended as a replacement for MOND phenomenology. It provides an interpretation of its low-acceleration successes as an emergent effective limit within a broader descriptive structure.

2.2. Effective Saturating Potential

The field equation introduced below is formally identical to the quasi-linear MOND formulation in the deep galactic regime. In the present work, however, this structure is not postulated as a fundamental modification of gravity. Instead, we adopt it as a minimal effective parametrization of a bounded-response principle governing low-density kinematic inference.

We define the effective gravitational potential Φ_{eff} through the nonlinear Poisson equation

$$\nabla \cdot \left[\mu \left(\frac{|\nabla \Phi_{\text{eff}}|}{a_*} \right) \nabla \Phi_{\text{eff}} \right] = 4\pi G \rho_b, \quad (1)$$

where ρ_b denotes the baryonic mass density. The function $\mu(x)$ encodes the progressive saturation of the effective response as the acceleration falls below the characteristic scale a_* .

For clarity, we work primarily with the μ -formulation in Eq. (1). An equivalent algebraic representation may be written in terms of an interpolation function $\nu(y)$ defined by $g_{\text{eff}} = \nu(y)g_N$ with $y = g_N/a_*$, where $\nu(y) \rightarrow 1$ for $y \gg 1$, ensuring an exactly Newtonian limit.

In the present interpretation, $\mu(x)$ does not represent a field-dependent coupling in the fundamental action. Rather, it parametrizes the smooth onset of a bounded-response regime in the effective description. The specific functional form adopted here provides a minimal interpolation between regimes and is not assumed to be unique.

The function $\mu(x)$ satisfies

$$\mu(x) \rightarrow 1 \quad \text{for } x \gg 1, \quad (2)$$

and

$$\mu(x) \rightarrow x \quad \text{for } x \ll 1, \quad (3)$$

ensuring recovery of Newtonian dynamics at high acceleration and asymptotically flat rotation curves at low acceleration.

For spherically symmetric systems, the effective radial acceleration satisfies

$$\mu\left(\frac{g_{\text{eff}}}{a_{\star}}\right)g_{\text{eff}} = g_{\text{N}}, \quad (4)$$

where g_{N} is sourced solely by baryons.

Equation (4) reproduces the observed flattening of galaxy rotation curves without introducing dark halos. The transition scale $a_{\star}$ controls both the onset of flattening and the asymptotic velocity amplitude.

Crucially, $a_{\star}$ is not treated as a tunable per-galaxy parameter. It is determined globally from the full rotation-curve sample (Section 3.4), and is interpreted here as the characteristic acceleration at which the effective geometric response enters a bounded regime.

2.3. Environmental Dependence and Low-Density Regimes

The effective modification introduced above is intrinsically sensitive to the gravitational environment. Regions of low baryonic density or weak gravitational gradients probe the saturated regime most strongly.

This environmental dependence becomes critical at cosmological scales. Cosmic voids, characterized by extreme matter dilution, naturally sample the deep saturation regime. As a result, local kinematic measurements performed in void-dominated regions may exhibit systematic offsets relative to globally inferred expansion rates.

Within this framework, the Hubble tension is not interpreted as a failure of early-universe cosmology. It arises as a late-time inference bias, reflecting the use of a single spacetime-based parametrization across environments characterized by different effective resolutions.

Galaxy rotation curves and the Hubble tension therefore emerge as two manifestations of the same low-density phenomenology. Galaxies probe saturation radially through declining surface density, while cosmic voids probe it volumetrically through large-scale dilution. The identification of a single scale $a_{\star}$ accounting for both effects constitutes the central organizing principle of this effective description.

The following sections test this effective framework against galactic rotation curve data and explore its implications for local measurements of the Hubble constant.

2.4. Operational Saturation Scale

The bounded-response behavior introduced above can be associated with an operational saturation scale. In the underlying relational description, the invariant bound b_{χ} limits the maximal rate at which admissible configurations can be updated. In the effective macroscopic regime, this constraint manifests as a characteristic acceleration proxy $a_{\star}$ marking the onset of non-linear response.

Dimensional consistency requires that this threshold be inversely proportional to the descriptive resolution $\ell(z)$ of the effective state U . We therefore parametrize

$$a_{\star}(z) \sim \kappa \frac{b_{\chi}}{\ell(z)}, \quad (5)$$

where κ is a dimensionless normalization factor encoding the convention adopted for the projective mapping. In the late-time low-density regime considered in this section, $a_{\star}$ may be treated as approximately constant. Its possible redshift dependence becomes relevant only in the primordial regime and will be discussed in Section 6.

2.5. Cosmological Descriptive Resolution

The parametrization $a_*(z) \sim \kappa b_\chi / \ell(z)$ requires specifying the descriptive resolution in the homogeneous regime.

In a spatially homogeneous late-time universe, no local preferred length scale exists. Operational closure therefore fixes $\ell(z)$ to the maximal domain over which an update of the projected state can coherently propagate within a single characteristic update time.

The only intrinsic timescale in the homogeneous regime is $\tau(z) \sim H(z)^{-1}$, which follows directly from $H = \dot{\chi}/\chi$. The maximal operational influence radius within one update step is $c \tau(z) = c/H(z)$.

Minimality of the closure then requires

$$\ell(z) = \frac{c}{H(z)}. \quad (6)$$

Substituting Eq. (6) into Eq. (5) gives

$$a_*(z) \sim \kappa c H(z). \quad (7)$$

At the present epoch this yields $a_*(t_0) \sim \kappa c H_0$, establishing a direct structural link between the galactic saturation scale and the cosmological expansion rate.

2.6. Holonomy Normalization of κ

The normalization factor κ can be fixed by consistency with the U(1) fiber structure established in the companion Born–Infeld analysis.

In the saturated regime, the effective connection amplitude is uniquely determined by scale invariance as

$$|A_T| \sim \frac{a_*}{c^2}. \quad (8)$$

Consider the canonical cosmological loop of circumference $L_H = 2\pi R_H = 2\pi c/H_0$. The accumulated holonomy is

$$\Delta\theta_H \sim \frac{a_*}{c^2} \cdot 2\pi R_H = \frac{2\pi a_*}{c H_0}. \quad (9)$$

Two intrinsic saturation criteria arise:

- Topological closure: $\Delta\theta_H = 2\pi$, giving $a_* = c H_0$.
- Coherence degradation: $\Delta\theta_H = 1$, giving $a_* = c H_0 / (2\pi)$.

The ratio between the two thresholds is fixed by the U(1) topology and is not adjusted.

3. Galaxy Rotation Curves

Throughout this section, the only galaxy-dependent free parameter is the stellar disc mass-to-light ratio Y_* , varied within a physically motivated interval. The saturation scale a_* is determined once from the full sample and then held fixed. No additional halo parameters are introduced.

We now test the effective model introduced in Section 2 against observed galaxy rotation curves. The goal of this section is not to achieve maximal fitting flexibility, but to assess whether a single effective saturation scale can reproduce the main kinematic features of disk galaxies using baryonic matter alone.

A useful internal consistency requirement is that the acceleration scale governing the transition be determined directly from galactic kinematic data. For the SPARC sample, fitting the effective model to the observed rotation curves yields a characteristic transition

scale of order $10^{-10} \text{ m s}^{-2}$. This value is obtained within the present framework solely from galaxy rotation data and does not rely on any cosmological prior or assumption regarding the Hubble constant.

The numerical value is consistent with the empirical acceleration scale identified in MOND analyses [2]. In the present framework, however, this scale is not introduced as a fundamental modification of the gravitational law. It emerges as the transition point of a bounded-response regime in the effective kinematic description.

This motivates treating a_* as a single universal effective scale consistently appearing in both high- and low-surface-density galaxies, without requiring modifications to early-time cosmology or additional dark components.

3.1. Data and Methodology

We use the SPARC database, which provides high-quality rotation curves together with homogeneous photometry and baryonic mass models for disk galaxies spanning a wide range of morphologies, surface brightnesses, and dynamical regimes [1].

The numerical pipeline used to generate the rotation-curve figures and diagnostics is available as open-source software [?].

For each galaxy, the baryonic mass distribution is constructed from the observed stellar and gas components. Stellar mass-to-light ratios are fixed following the standard SPARC prescriptions. No additional dark matter component is introduced in this effective description.

The effective acceleration $g_{\text{eff}}(r)$ is obtained by solving Eq. (4) using the Newtonian acceleration $g_{\text{N}}(r)$ computed from the baryonic mass distribution. The circular velocity then follows from

$$v_{\text{eff}}(r) = \sqrt{r g_{\text{eff}}(r)}. \quad (10)$$

The saturation scale a_* is treated as a universal parameter. Its value is fixed globally by minimizing the total residuals across the sample, rather than optimized on a galaxy-by-galaxy basis.

3.2. Representative Fits

We illustrate the performance of the effective model on representative galaxies spanning different surface-density regimes. The goal is not to emphasize isolated best cases, but to show typical behaviors observed across the full SPARC sample.

High-surface-density galaxies are well described by Newtonian dynamics in their inner regions. Deviations from Newtonian predictions appear only at large radii, where the baryonic acceleration drops below the saturation scale. The resulting rotation curves flatten smoothly, without requiring extended mass halos, in agreement with observed trends across high-surface-brightness disks [1].

Low-surface-brightness galaxies probe the saturated regime over most of their radial extent. In these systems, the effective acceleration departs from Newtonian behavior already at small radii, naturally producing slowly rising and asymptotically flat rotation curves, a feature commonly observed in diffuse systems [2].

Across the sample, the model reproduces the observed diversity of rotation curve shapes. This diversity arises solely from differences in the baryonic mass distribution and surface density, not from variations in the effective parameters.

The galaxies shown in Fig. 1 are selected as representative well-fitted systems spanning rising, nearly flat, and slowly declining rotation curves. The saturation scale a_* is fixed globally from the full-sample analysis discussed in Section 3.4. No galaxy-dependent adjustment of a_* is performed.

Representative examples illustrating these behaviors are shown in Fig. 1.

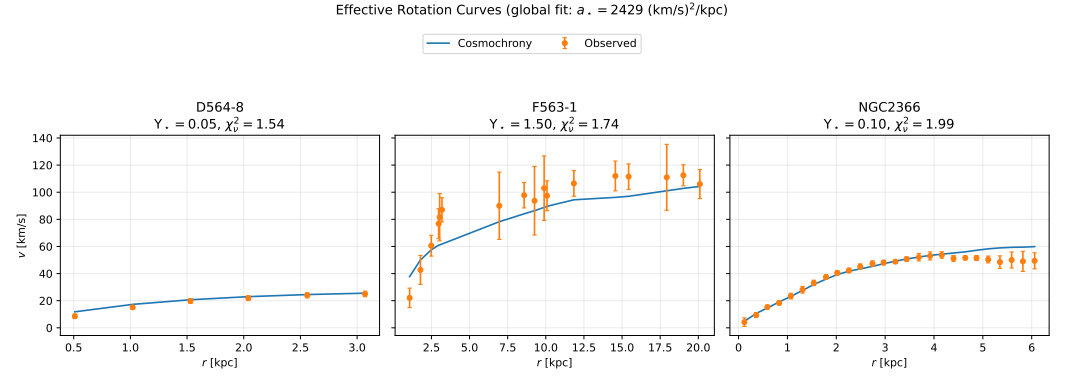


Figure 1. Representative galaxy rotation curves from the SPARC sample. Observed circular velocities (points) are compared with the effective model prediction (solid lines) computed from the baryonic mass distribution alone. The three systems span distinct surface-density regimes and kinematic behaviors (rising, nearly flat, and mildly declining). A single universal saturation scale a_* , determined from the full sample, is used for all galaxies. The only galaxy-dependent parameter is the stellar mass-to-light ratio Y_* , fitted within a physically motivated interval. The code used to generate these curves is publicly available [?].

3.3. Residuals and Scaling Relations

We quantify the quality of the fits using radial velocity residuals

$$\Delta v(r) = v_{\text{obs}}(r) - v_{\text{eff}}(r). \quad (11)$$

The residuals exhibit no systematic radial trends across the sample. In particular, no characteristic scale or radius-dependent bias is observed.

The effective model naturally reproduces the empirical correlation between rotation curve shape and baryonic surface density. Galaxies with higher central surface density remain in the Newtonian regime over a larger radial range, while diffuse systems enter the saturated regime earlier.

The transition radius at which rotation curves flatten corresponds closely to the regime where the effective surface flux density approaches the saturation limit. Beyond this point, further decreases in baryonic density no longer translate into stronger dynamical suppression, leading to asymptotically flat rotation velocities.

This behavior is closely related to the observed radial acceleration relation. In the present framework, this relation is not imposed but emerges directly from the effective saturation of the gravitational response at low acceleration [2].

3.4. Universality of the Saturation Scale

A central outcome of the analysis is the empirical stability of the saturation scale a_* when determined from the full SPARC sample ($N = 175$ galaxies). The scale is obtained through a global robust optimization, varying only the stellar disc mass-to-light ratio Y_* on a galaxy-by-galaxy basis, while keeping Y_{bul} fixed.

Using a median-based estimator and allowing $Y_* \in [0.01, 2.0]$, we obtain $a_* \simeq 2.4 \times 10^3 (\text{km s}^{-1})^2 \text{kpc}^{-1}$. A trimmed-mean criterion yields $a_* \simeq 2.8 \times 10^3 (\text{km s}^{-1})^2 \text{kpc}^{-1}$. The inferred scale is therefore stable at the $\sim 15\%$ level with respect to the choice of robust estimator.

The resulting distribution of reduced χ^2 values is shown in Fig. 2. The median value is $\chi_v^2 \approx 2.7$, with approximately 60–65% of galaxies satisfying $\chi_v^2 < 5$ and nearly 80% satisfying $\chi_v^2 < 10$. These values are obtained without introducing galaxy-dependent halo parameters or cosmological priors.

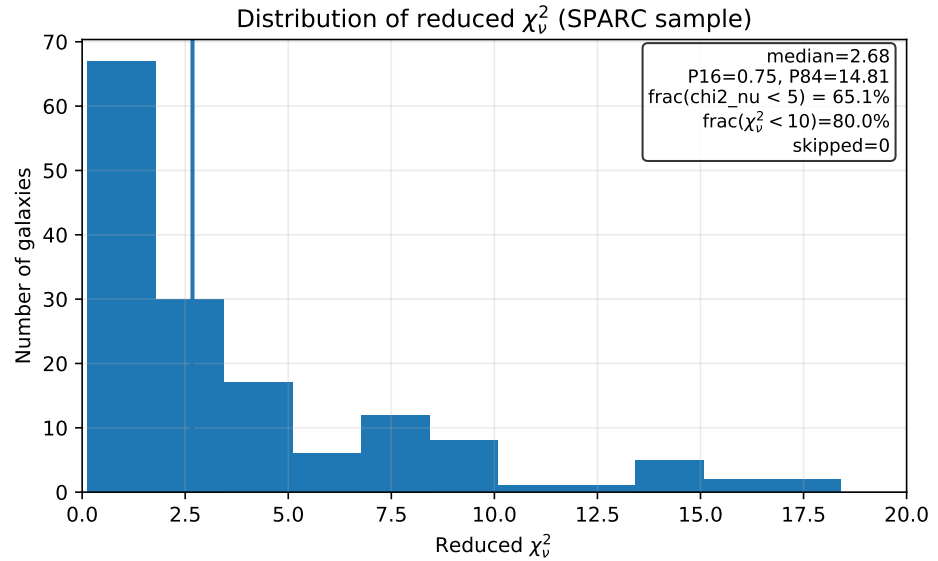


Figure 2. Distribution of reduced χ^2 values for the full SPARC sample ($N = 175$ galaxies) using the globally fitted saturation scale a_* . The vertical line indicates the median value.

The inferred scale remains within the same range when adopting plausible variations of the bulge mass-to-light ratio ($Y_{bul} = 0.3\text{--}0.7$), indicating that a_* is not driven by fine-tuning of baryonic normalization.

This supports the interpretation of a_* as a universal effective transition scale rather than as a phenomenological fitting parameter. All galaxy-to-galaxy variability is accounted for by the observed baryonic structure, with no halo-by-halo adjustment.

In the present framework, the universality of the saturation scale reflects the existence of a bounded-response regime in the effective geometric description, rather than a modification of the fundamental gravitational law.

The statistical performance obtained here is comparable to that reported in published MOND fits to the same SPARC dataset, but is derived from an effective saturation principle rather than a postulated modification of the gravitational field equation.

The absence of a secondary high- χ^2 population suggests that the model does not rely on a fine-tuned subset of galaxies.

3.5. Physical Interpretation of the Saturation Amplitude

The parameter $f_{\max}$ does not represent an adjustable fit coefficient, but the maximal admissible fractional deficit compatible with the bounded-response structure defined by Eq. (1).

Consistency with galactic dynamics constrains its range. Values $f_{\max} \ll 0.05$ fail to reproduce the observed MOND-like transition in low-surface-brightness galaxies, while values $f_{\max} \gtrsim 0.15$ would induce measurable departures from Newtonian behavior in high-acceleration environments.

The admissible interval $0.05 \lesssim f_{\max} \lesssim 0.15$ is therefore fixed by internal cross-scale consistency, rather than by cosmological tuning.

Within this range, the void-induced fractional shift $\Delta H/H \simeq \delta_\rho f_{\max}$ naturally falls at the percent level for observed void contrasts $\delta_\rho \sim 0.2\text{--}0.4$, without requiring additional degrees of freedom.

3.6. Summary

The effective saturation model reproduces the main phenomenological features of galaxy rotation curves across a wide range of systems using baryonic matter alone. The

same universal saturation scale governs both high- and low-surface-density galaxies, with smooth transitions between Newtonian and saturated regimes.

These results motivate extending the analysis to cosmological low-density environments. In the next section, we show that the same effective mechanism leads to environment-dependent signatures in the inferred expansion rate, providing a natural connection to the Hubble tension.

The determination of a_* from galactic rotation curves constitutes the only free parameter of the effective model. No cosmological quantity, including H_0 , enters this calibration. The cosmological implications discussed in Section 4 therefore represent an independent consequence of the same empirically fixed scale.

Importantly, the calibration of a_* uses only galactic kinematic data and does not involve any cosmological observable.

4. Cosmic Voids and the Hubble Tension

In this section we explore the cosmological implications of the effective saturation mechanism introduced above. We focus on late-time, low-density environments and show how the same effective transition structure responsible for flat galaxy rotation curves naturally leads to environment-dependent signatures in the locally inferred expansion rate.

The analysis is restricted to late-time phenomenology at low redshift. Early-universe physics and the global expansion history are assumed to remain unchanged.

It is important to clarify that the cosmological argument developed in this section does not follow from standard MOND phenomenology alone. In conventional MOND frameworks, the modified field equation is treated as a fundamental alteration of gravity at galactic scales, without generically implying an environment-dependent bias in the locally inferred Hubble parameter. Here, by contrast, the same bounded-response principle that governs galactic dynamics extends consistently to volumetric dilution in cosmic voids, linking rotation curves and Hubble inference within a single effective saturation mechanism.

Importantly, the saturation scale a_* entering the void analysis is fixed entirely by galaxy rotation curve data (Section 3). No cosmological parameter, including the Hubble constant, enters the determination of a_* . The void analysis therefore does not tune H_0 , but explores the consequences of a scale already fixed by independent galactic observations.

The background Friedmann evolution remains strictly unchanged. The mechanism discussed here does not alter the homogeneous expansion history, but affects the local kinematic inference of the expansion rate in strongly underdense environments.

Unlike conventional local-void explanations of the Hubble tension, which require a large-scale density fluctuation tuned to specific amplitudes and spatial profiles, the present mechanism follows directly from the same bounded-response principle that governs galactic dynamics. The effect is therefore not introduced ad hoc, but arises from a scale already constrained by independent data.

4.1. Structural Unification of the Kinematic Deficit

The transition from galactic scales to cosmological expansion is not a change in the gravitational law but a property of the effective inference metric itself.

Equation (1) expresses a bounded-response relation between the Newtonian source g_N and the inferred effective acceleration g_{eff} . In the deep saturation regime, the response function $\mu(x)$ enforces

$$g_{\text{eff}} \propto \sqrt{a_* g_N},$$

which limits the admissible growth of the inferred acceleration when $g_N \ll a_*$.

This limitation is not tied to radial geometry. It reflects a bound on the norm of the inferred gradient of the effective potential,

$$\|\nabla\Phi_{\text{eff}}\| \leq a_*,$$

which applies independently of whether the dilution is radial (galactic outskirts) or volumetric (cosmic voids).

In galactic systems, decreasing surface density drives g_N below a_* , producing flat rotation curves as shown in Section 3. In cosmological voids, decreasing volumetric density suppresses the effective gradient of the large-scale metric inference, leading to a bounded deficit in the locally inferred expansion rate.

The projection deficit δ entering Eq. (8) is therefore the volumetric analogue of the radial deficit encoded in Eq. (4). Both emerge from the same bounded-response constraint on the effective metric.

The relationship between Eq. (1) and Eq. (8) is thus mediated by a scale-invariant inference threshold a_* , which remains constant across geometries.

This derivation assumes that the saturation effect acts at the level of geometric inference during kinematic reconstruction, rather than altering the fundamental growth equation or the null geodesics of light propagation.

4.2. Local Expansion in Low-Density Environments

Measurements of the Hubble constant rely on local kinematic observables, including distance ladders, standard candles, and redshift–distance relations. These measurements implicitly assume that the locally inferred expansion rate is representative of the global cosmological background [5,11].

However, observations of large-scale structure show that the late-time Universe is strongly inhomogeneous. A substantial fraction of the cosmic volume is occupied by voids, characterized by matter densities well below the cosmic mean [12].

Within the effective framework introduced in Section 2, low-density environments probe the saturated regime of the effective response. In this regime, local kinematic observables no longer sample the same effective transition resolution as in high-density regions. As a result, expansion rates inferred from such environments need not coincide with the global average, even in the absence of any modification of early-time cosmology.

We therefore distinguish between a global expansion rate H_{global} , defined by the homogeneous background constrained by early-universe observables [6], and an effective local expansion rate H_{loc} , inferred from observations within a specific late-time environment.

4.3. Effective Expansion Rate in Voids

Cosmic voids represent the deepest and most extended low-density regions in the late Universe. Within voids, baryonic and total matter densities are strongly suppressed, and typical gravitational accelerations fall below the saturation scale a_* over large volumes [12].

In this regime, the effective description operates close to its low-density saturation limit. The relation between local kinematics and the underlying background expansion is therefore modified at the level of inference rather than at the level of the dynamical Friedmann equations. To leading order, this effect can be captured by an environment-dependent renormalization of the inferred expansion rate,

$$H_{\text{loc}} = H_{\text{global}}[1 + \delta_{\text{eff}}], \quad \text{with} \quad \delta_{\text{eff}} = \delta_\rho f\left(\frac{g_{\text{void}}}{a_*}\right), \quad (12)$$

where $\delta_\rho = 1 - \rho_{\text{void}}/\rho_{\text{global}}$ denotes the density contrast of the void, and g_{void} represents a characteristic gravitational acceleration scale associated with the void, estimated from

its typical size and density contrast. This expression does not modify the background Friedmann equations. It parametrizes a bias in the kinematic inference of the expansion rate in underdense environments.

The function $f(x)$ is a dimensionless saturation function determined by the same bounded-response structure that governs galactic dynamics. In the deep saturated regime ($x \ll 1$), $f(x)$ approaches a finite asymptotic value $f_{\max}$, set by the maximal projection deficit allowed by the effective description. Independent constraints derived in Section 4.3 of [8] imply $f_{\max} \lesssim 0.1$.

The saturation scale a_* entering Eq. (12) is fixed independently from galactic rotation curves (Section 3.4), and no cosmological parameter enters its determination.

For void density contrasts in the observationally inferred range $\delta_\rho \sim 0.2\text{--}0.4$ [12], and for $g_{\text{void}} \ll a_*$, Eq. (12) yields a characteristic fractional shift $\Delta H/H$ at the few-percent level. This magnitude is comparable to the observed local offset, without requiring any tuning of cosmological parameters.

The magnitude of the effect therefore follows directly from combining (i) the empirically determined scale a_* , and (ii) independently measured void density contrasts.

The correction δ_{eff} depends on the depth and spatial extent of the void, as well as on the empirically determined saturation scale a_* . Denser or more weakly underdense regions remain close to the unsaturated regime and yield $\delta_{\text{eff}} \approx 0$.

Importantly, this mechanism does not introduce any additional energy component and does not modify the background Friedmann equations. The global expansion history remains unchanged. Only the local kinematic inference of the expansion rate is affected.

Interpretational scope.

The bounded-response mechanism discussed here does not alter the fundamental dynamical equations governing matter evolution, nor does it modify null geodesics of photon propagation. The continuity equation and the standard linear growth rate $f_g(\Omega_m)$ remain unchanged. The bias δ_{eff} therefore arises at the level of coarse-grained kinematic reconstruction of the expansion rate in underdense environments, rather than from a modification of local gravitational dynamics.

Order-of-magnitude estimate.

A minimal numerical closure follows from estimating the characteristic void acceleration at radius R_{void} from the enclosed mass deficit,

$$g_{\text{void}} \sim \frac{4\pi}{3} G \rho_m \delta_\rho R_{\text{void}}, \quad (13)$$

with $\rho_m = 3H_0^2 \Omega_m / (8\pi G)$, hence

$$g_{\text{void}} \sim \frac{1}{2} \Omega_m H_0^2 \delta_\rho R_{\text{void}}. \quad (14)$$

Taking a representative large void $R_{\text{void}} \sim 100\text{--}200$ Mpc and $\delta_\rho \sim 0.2\text{--}0.4$ [12] gives $g_{\text{void}} \sim (0.5\text{--}2) \times 10^{-12} \text{ m s}^{-2}$. The saturation scale fixed by rotation curves is $a_* \simeq 2.4 \times 10^3 (\text{km s}^{-1})^2 \text{ kpc}^{-1}$ (Section 3.4), i.e. $a_* \simeq 7.8 \times 10^{-11} \text{ m s}^{-2}$, so $g_{\text{void}}/a_* \lesssim 3 \times 10^{-2} \ll 1$. The deep-saturation limit therefore applies and Eq. (12) reduces to

$$\frac{\Delta H}{H} \equiv \delta_{\text{eff}} \simeq \delta_\rho f_{\max}, \quad (15)$$

yielding $\Delta H/H \sim 0.01\text{--}0.04$ for $\delta_\rho \sim 0.2\text{--}0.4$ and $f_{\max} \lesssim 0.1$.

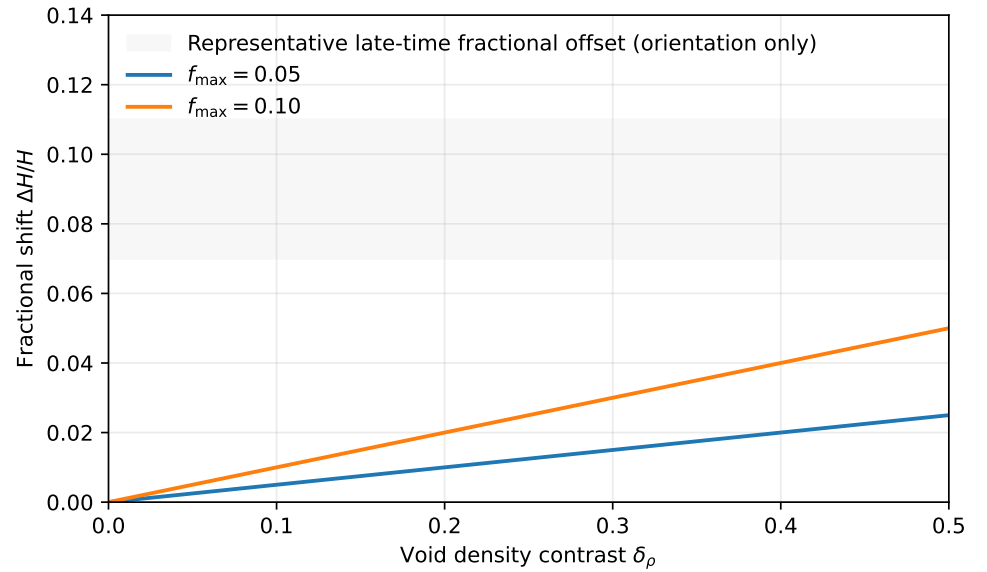


Figure 3. Quantitative visualization of the void inference bias implied by Eq. (12) in the deep-saturation regime ($g_{\text{void}} \ll a_*$), where $f(g_{\text{void}}/a_*) \rightarrow f_{\max}$ and $\Delta H/H \simeq \delta_\rho f_{\max}$ (Eq. (15)). Curves show representative values $f_{\max} = 0.05$ and $f_{\max} = 0.10$, consistent with the independent bound discussed in the text. The shaded horizontal band indicates a representative range of late-time fractional offsets reported in the literature, shown for orientation only and not used for calibration.

For orientation, we indicate as a shaded benchmark band the typical fractional offset discussed in the Hubble-constant literature [5–7].

Within this interpretation, the enhanced expansion inferred in cosmic voids reflects a reduced effective geometric constraint rather than an additional energy density. In regions where the density of contributing relational structure is low, the kinematic inference based on sparse mass distributions operates closer to its saturation limit, inducing a systematic environment-dependent bias in locally inferred kinematic relations.

4.4. Connection to the Hubble Tension

The Hubble tension refers to the discrepancy between early-time determinations of the Hubble constant, primarily inferred from cosmic microwave background measurements, and late-time local measurements based on the distance ladder [5,6]. Despite extensive investigation, no single explanation has yet achieved broad consensus [7].

In the present framework, this discrepancy is interpreted as a manifestation of environment-dependent kinematic inference. Early-time measurements are anchored in the nearly homogeneous, high-density Universe, where the effective description operates in the unsaturated regime and recovers H_{global} . By contrast, late-time distance-ladder measurements are performed in a highly structured Universe and may preferentially probe void-dominated environments.

If observers and calibrator systems reside in regions operating near the saturated regime, Eq. (12) predicts a systematic upward shift in the locally inferred expansion rate,

$$H_{\text{loc}} > H_{\text{global}}. \quad (16)$$

The magnitude of the shift depends on the depth and spatial extent of the surrounding underdensity.

The quantitative dependence of the fractional shift on the void density contrast, in the deep-saturation regime fixed by a_* , is summarized in Fig. 3, which directly visualizes Eq. (15) without introducing any additional cosmological parameter.

Importantly, this mechanism does not alter the background Friedmann evolution. It does not introduce new relativistic degrees of freedom, nor does it modify early-time cosmology. The effect arises solely from the bounded-response structure governing late-time low-density environments.

The extent to which local underdensities alone can account for the full observed tension remains debated [13,14]. The present framework does not posit a single finely tuned large-scale void. Instead, it provides a distinct mechanism in which the same empirically determined saturation scale a_* that governs galactic dynamics induces a systematic environment-dependent bias in late-time kinematic inference.

The proposal is therefore not a local-void cosmology, but a unified effective description in which both galaxy rotation curves and the Hubble offset arise from a common bounded-response principle.

Because the dynamics derives from an effective potential Φ_{eff} , the framework admits a conserved effective energy and a corresponding modified virial relation for stationary systems. This ensures that equilibrium disk galaxies remain dynamically stable in the transition regime and that the void mechanism does not introduce inconsistencies at galactic scales.

Early-universe inferences (CMB) probe a nearly homogeneous high-density regime well above the saturation threshold, and therefore recover H_{global} .

4.5. Observable Signatures

The effective saturation framework leads to several testable observational signatures that distinguish it from both Λ CDM and purely local-void interpretations.

First, the locally inferred Hubble constant should correlate with the large-scale density environment. Measurements performed in or near deep voids are expected to yield systematically higher values of H_0 than those obtained in denser regions. This correlation should persist after controlling for known observational systematics and distance-calibration effects.

Second, the effect should exhibit a characteristic redshift dependence. At sufficiently large redshift, line-of-sight averaging over multiple environments suppresses the environmental bias. The inferred expansion rate should therefore converge toward H_{global} , producing a gradual reduction of the offset with increasing survey depth.

Third, the framework predicts environment-dependent signatures in void-related observables, including weak lensing profiles and redshift drift measurements [15]. In particular, if the effective expansion rate inside voids is enhanced relative to the global value, the temporal evolution of redshift for sources embedded in deep underdensities should deviate from the homogeneous expectation in a systematic way.

More generally, the predicted shift depends not only on the local environment of the observer, but on the integrated low-density structure sampled along the line of sight. Both the location of the source and the geometry of intervening voids contribute to the inferred bias. This implies that samples with different sky coverage or different environmental selection criteria may yield systematically different H_0 estimates.

Importantly, the framework does not predict a unique numerical value for the offset in a given environment. Instead, once a_* is fixed independently from galactic rotation curves, the sign, scaling behavior, and environmental dependence of the effect are determined. This renders the proposal falsifiable with current and forthcoming large-scale-structure and distance-ladder data.

4.6. Predicted Amplitude and Redshift Dependence

A key feature of the present framework is that the magnitude of the locally inferred deviation from the global expansion rate is constrained by the saturation scale a_* , which is fixed independently from galaxy rotation curves.

For representative underdensities characteristic of large cosmic voids on scales of a few hundred megaparsecs, and for $g_{\text{void}} \ll a_*$, the framework predicts a fractional enhancement

$$\frac{\Delta H}{H} \equiv \frac{H_{\text{loc}} - H_{\text{global}}}{H_{\text{global}}} \sim \text{a few percent}, \quad (17)$$

with the precise value controlled by the empirically determined scale a_* and by independently measured void density contrasts.

The effect is intrinsically redshift dependent. At higher redshift, large-scale structure is less developed, density contrasts are reduced, and line-of-sight averaging over multiple environments becomes increasingly efficient. Consequently, the locally inferred expansion rate is expected to converge progressively toward H_{global} with increasing redshift.

The framework therefore predicts a smooth decrease of $\Delta H/H$ as a function of survey depth. A persistent offset at high redshift, inconsistent with environmental averaging, would falsify the mechanism.

Conversely, deviations substantially larger than the percent-to-few-percent range, or the complete absence of any measurable environment-dependent trend once systematics are controlled, would directly challenge the proposed interpretation.

4.7. Interpretation of the Hubble Tension as an Environmental Effect

The framework developed here suggests that the Hubble tension does not reflect a change in the global expansion history. It reflects a limitation of homogeneous kinematic inference when applied to a strongly inhomogeneous late-time Universe.

Early-universe determinations of H_0 , anchored in a nearly homogeneous background, recover the global expansion rate. By contrast, late-time local measurements probe structured environments in which the effective transition resolution operates in a low-density saturated regime.

In this context, the discrepancy between early- and late-time determinations of H_0 arises from a mismatch between global averaging and local inference. Measurements performed in void-dominated regions sample an effective response that no longer scales linearly with decreasing density, leading to a systematically enhanced locally inferred expansion rate.

From this perspective, the Hubble tension is not an anomaly requiring additional dark components or early-time modifications. It is a diagnostic of gravitational and kinematic inference in diffuse environments. This interpretation naturally predicts weak but systematic correlations between inferred values of H_0 and large-scale environmental indicators such as void depth or filament membership.

4.8. Summary

In this section we have shown that the same effective saturation mechanism that accounts for galaxy rotation curves naturally extends to late-time, low-density cosmological environments.

When the characteristic gravitational field falls below the empirically determined scale a_* , the effective description operates in a bounded-response regime. In cosmic voids, this regime induces an environment-dependent bias in the locally inferred expansion rate, while leaving the global background dynamics unchanged.

The mechanism therefore provides a structural link between galactic kinematics and the Hubble tension: both phenomena arise from the same transition scale, independently fixed by rotation curve data.

The predicted amplitude of the effect is constrained to the percent-to-few-percent level, depends on independently measured void density contrasts, and decreases with increasing redshift due to environmental averaging. These features render the proposal observationally testable.

In the next section, we examine the robustness of these results, discuss degeneracies with standard cosmological effects, and clarify the domain of validity of the effective description.

5. Robustness, Degeneracies, and Limits

In this section we assess the robustness of the effective saturation framework, its potential degeneracies with standard astrophysical and cosmological effects, and the limits of its domain of applicability. This analysis is essential to determine whether the proposed mechanism constitutes a genuine explanatory alternative or merely a reparameterization of existing descriptions.

5.1. Baryonic Uncertainties

The predictions of the effective framework depend explicitly on the observed baryonic mass distribution. Uncertainties in stellar mass-to-light ratios, gas content, and distance estimates therefore propagate into the inferred rotation curves. Such uncertainties are well documented and represent a generic limitation of baryon-based dynamical modeling [1].

We have tested the sensitivity of the effective fits to reasonable variations in stellar mass normalization. While moderate shifts in mass-to-light ratio affect the detailed shape of the inner rotation curves, they do not eliminate the emergence of a low-acceleration saturation regime at large radii. In particular, the appearance of asymptotically flat rotation curves remains robust across the allowed baryonic parameter range, consistent with empirical scaling relations [2].

At cosmological scales, baryonic uncertainties primarily affect the mapping between large-scale structure and local observational samples. They do not remove the qualitative distinction between dense environments, which probe an unsaturated effective regime, and cosmic voids, which sample the saturated regime most strongly [12].

5.2. Conceptual Position within Modified-Gravity and Inhomogeneous Frameworks

The phenomenology discussed in this work exhibits partial overlap with acceleration-based approaches such as Modified Newtonian Dynamics (MOND), and the cosmological discussion involves large-scale underdensities. It is therefore important to clarify the conceptual position of the present framework within this broader landscape.

Relation to MOND-like phenomenology.

At galactic accelerations below a_* , Eq. (1) becomes formally equivalent to quasi-linear MOND in the weak-field regime. Both approaches identify a characteristic acceleration scale below which Newtonian expectations break down.

The distinction is structural rather than mathematical. In MOND, the acceleration scale a_0 is introduced as a fundamental modification of the gravitational law. In the present framework, the same scaling behavior arises as the low-density limit of a bounded-response principle. The scale a_* characterizes the onset of saturation in the effective description, not a new force law.

Crucially, the extension to cosmological inference does not follow from standard MOND formulations. Here, both galactic dynamics and the environment-dependent

Hubble offset derive from the same empirically calibrated scale a_* , linking two regimes that are usually treated independently.

Distinction from inhomogeneous cosmologies.

Models based on large-scale metric inhomogeneities, such as Lemaître–Tolman–Bondi (LTB) constructions, attempt to account for late-time discrepancies by modifying the background geometry itself. In such scenarios, the Hubble parameter becomes position dependent because the spacetime metric deviates from the homogeneous Friedmann–Lemaître–Robertson–Walker solution.

The present framework does not modify the background metric and does not introduce a position-dependent Friedmann solution. The global expansion history remains that of standard cosmology. The predicted offset arises instead from an environment-dependent modification of the effective kinematic inference once local accelerations fall below the saturation scale a_* .

The mechanism therefore does not require a special observer location or a finely tuned void profile. The shift in the locally inferred expansion rate scales with the surrounding density contrast according to Eq. (12), while preserving global homogeneity at the level of the background solution.

Structural distinction.

The unifying feature of the framework is the existence of a universal bound on effective response. This bound is not environment dependent. What varies across environments is the density of relations contributing to the effective projection.

Dense systems remain in the unsaturated regime, while extended low-density regions approach saturation. The resulting correlation between galactic kinematics and late-time Hubble inference is therefore not a reparameterization of dark sector physics nor a modification of the cosmological metric, but a manifestation of a single bounded-response structure operating across scales.

5.3. Degeneracies with Standard Astrophysical Effects

Any proposed alternative mechanism must be carefully examined against established astrophysical and cosmological effects within the Λ CDM framework. We therefore distinguish between degeneracies at galactic and cosmological scales.

Galactic scales.

Within dark matter halo models, baryonic feedback processes and baryon–halo coupling have been shown to modify inner halo profiles and reproduce certain features of flattened rotation curves, particularly in low-mass systems [16]. Such models typically introduce galaxy-dependent halo parameters and rely on a dynamical coupling between baryonic surface density and halo response.

By contrast, the effective saturation framework introduces no halo degrees of freedom and involves a single universal acceleration scale a_* . All galaxy-to-galaxy diversity arises solely from the observed baryonic mass distribution and from the fitted stellar mass-to-light ratio. The resulting scaling relations follow directly from the structure of Eq. (1), without requiring halo-by-halo tuning or environment-dependent structural parameters.

A degeneracy may therefore exist at the level of rotation-curve fitting accuracy. However, the underlying parameter structure differs qualitatively: halo models accommodate diversity through adjustable halo profiles, whereas the present framework attributes diversity to variations in baryonic structure under a fixed saturation scale. This difference implies distinct residual patterns and scaling correlations, providing a concrete avenue for empirical discrimination in detailed rotation-curve analyses.

Cosmological scales.

At low redshift, local inhomogeneities, peculiar velocities, and sample variance are known to affect distance-ladder determinations of the Hubble constant [11]. Large-scale underdensities have also been explored as partial explanations of the Hubble tension [13, 14]. In most such analyses, these effects are treated as perturbations of the background expansion within standard gravity.

The present framework differs in a crucial respect. The background Friedmann expansion remains unmodified. The predicted offset arises not from a change in global dynamics, nor from a radially tuned LTB-like void profile, but from an environment-dependent modification of effective kinematic inference once the local acceleration falls below the saturation scale a_* .

In particular, the mechanism does not rely on a large violation of homogeneity or on fine-tuned radial density profiles. Instead, it predicts a systematic scaling of the inferred expansion rate with the local density contrast, as encoded in Eq. (12). This scaling is tied to the same universal parameter a_* determined independently from galaxy rotation curves.

Conceptual distinction.

A central ingredient of the framework is the existence of a universal bound on the effective response of the geometric description. This bound is not environment dependent. What varies across environments is the density of relations contributing to the effective projection.

Dense systems concentrate many effective relations within limited volumes, keeping the dynamics close to the unsaturated regime. Extended low-density regions distribute the same fixed descriptive resolution over much larger volumes, pushing the effective description closer to saturation. The resulting enhancement of the locally inferred expansion rate therefore reflects how different environments sample a fixed bounded-response structure, rather than a modification of gravity, a change in the Friedmann background, or a reparameterization of dark-sector physics.

This structural distinction cannot be absorbed into standard astrophysical degeneracies. It predicts correlated galactic and cosmological signatures tied to a single universal scale, providing a clear and testable avenue for empirical discrimination.

5.4. Environmental Selection Effects

Local measurements of the Hubble constant are not uniformly distributed across cosmic environments. Distance ladder calibrators and standard candles are preferentially observed in specific regions of the large-scale structure [5].

This selection effect plays a central role in the present framework. If observational strategies preferentially sample void-dominated or underdense regions, the inferred expansion rate is expected to be systematically enhanced relative to the global value.

Importantly, this effect is not introduced as an ad hoc assumption. It constitutes a direct and testable prediction. Future analyses correlating inferred values of H_0 with void catalogs and density reconstructions can directly assess the magnitude of the effect, as already explored in the context of local underdensity scenarios [13,14].

5.5. Range of Validity

The effective saturation framework is not intended to apply universally at all scales and epochs. Its domain of validity is restricted to late-time, low-density environments.

In high-density regimes, including the early Universe, galaxy interiors, and the Solar System, the effective description reduces to standard Newtonian and relativistic behavior

by construction. No deviations from established physics are expected in these contexts, consistent with precision tests of gravity [17].

At sufficiently large redshift, line-of-sight averaging over multiple environments suppresses the saturation signature. The framework therefore predicts convergence toward standard cosmological behavior at high redshift, in agreement with current observations [6].

5.6. Falsifiability

The effective saturation framework makes a number of concrete, observationally testable predictions. Because the same universal scale a_* governs both galactic dynamics and late-time kinematic inference, the model can be falsified at multiple, independent levels.

(i) Absence of environment dependence in H_0 .

The framework predicts a correlation between the locally inferred Hubble parameter and the surrounding large-scale density contrast, as expressed in Eq. (12). Environment-stratified analyses of distance-ladder measurements should reveal a statistically significant trend between inferred H_0 and void proximity or density contrast. If no such correlation is detected within observational precision, or if the inferred H_0 proves statistically independent of environment, the proposed mechanism would be strongly disfavored.

(ii) Redshift scaling of the offset.

The predicted deviation $\Delta H/H$ is expected to decrease with increasing redshift due to environmental averaging along extended lines of sight. A persistent offset of comparable magnitude at higher redshift, inconsistent with large-scale structure dilution, would contradict the core mechanism. Conversely, a measured redshift scaling incompatible with the functional dependence implied by Eq. (12) would also falsify the framework.

(iii) Inconsistency of the universal scale.

The acceleration scale a_* is fixed from galaxy rotation curves. Cosmological applications do not introduce a new free parameter. If independent cosmological analyses required a significantly different effective scale to reproduce any environment-dependent expansion signal, the internal consistency of the framework would be violated. The model therefore predicts cross-consistency between galactic and cosmological determinations of the same parameter.

(iv) Breakdown of galactic saturation behavior.

The model predicts systematic low-acceleration behavior tied to a single universal scale across high- and low-surface-density galaxies. A statistically significant population of extremely diffuse systems that cannot be reconciled with a single a_* , or that require environment-dependent tuning of the scale, would invalidate the bounded-response interpretation.

(v) Standard-model sufficiency.

Finally, if improved Λ CDM-based modeling of feedback, selection effects, and local structure fully resolves both galaxy-scale phenomenology and the Hubble tension without invoking any environment-dependent scaling tied to a universal acceleration threshold, the explanatory necessity of the present mechanism would disappear.

These criteria render the framework empirically vulnerable. It does not merely accommodate the observed anomalies, but constrains their amplitude, scaling, redshift

dependence, and cross-scale consistency. Failure of any of the above conditions would require revision or rejection of the proposed mechanism.

5.7. Summary

The effective saturation framework remains stable under reasonable variations of baryonic mass-to-light ratios and does not rely on galaxy-dependent halo parameters. The acceleration scale a_* is determined directly from rotation curves and subsequently applied, without readjustment, to late-time low-density environments.

At cosmological scales, the mechanism does not modify the background Friedmann expansion. The predicted offset in locally inferred H_0 arises from an environment-dependent saturation of the effective kinematic inference, rather than from additional energy components or altered gravitational dynamics.

The proposal is therefore both constrained and falsifiable. Its viability depends on the simultaneous validity of: (i) a universal saturation scale across disk galaxies, and (ii) a measurable correlation between locally inferred expansion rates and large-scale density contrast. Failure of either condition would invalidate the framework.

6. Predictions and Observational Tests

The effective saturation framework introduced in this work leads to a set of distinct and testable predictions. These predictions follow directly from the environmental dependence of the effective transition resolution and do not rely on additional assumptions or parameter tuning.

6.1. Environment-Dependent Hubble Measurements

A primary prediction of the framework is a correlation between the locally inferred Hubble constant and the surrounding large-scale environment.

Observers and standard candles located in or near deep cosmic voids are expected to yield systematically higher inferred values of H_0 than those located in denser regions. Conversely, measurements anchored in overdense environments should converge more closely toward the global expansion rate H_{global} .

This prediction can be tested by cross-correlating existing and forthcoming H_0 measurements with void catalogs and density reconstructions derived from large-scale galaxy surveys, as already explored in the context of local underdensity scenarios [12–14]. A statistically significant null result would strongly disfavor the effective saturation mechanism.

In particular, the framework predicts a differential signal: measurements performed in environments of comparable redshift but contrasting large-scale density should exhibit systematically different inferred values of H_0 .

6.2. Redshift Dependence of the Hubble Offset

The environmental bias predicted by the framework is inherently scale dependent.

At sufficiently low redshift, local measurements are sensitive to individual large-scale structures and directly probe the saturated regime. At increasing redshift, line-of-sight averaging over multiple environments progressively suppresses the bias.

As a result, the inferred expansion rate is predicted to converge smoothly toward H_{global} with increasing redshift.

This behavior provides a clear observational signature. Measurements of the Hubble constant performed over different redshift ranges should exhibit a systematic trend, with the largest deviations appearing at the lowest redshifts, followed by a gradual convergence at higher redshift [7].

An important quantitative consequence of the framework is that the expected deviation remains at the percent level. Once the saturation scale a_* is fixed by galactic dynamics, the

fractional offset in the inferred expansion rate is predicted to peak at intermediate redshift and to remain negligible at both very low and sufficiently high redshift.

The peak is expected when the typical line-of-sight length becomes comparable to the characteristic scale of large voids, before full environmental averaging sets in.

Such deviations fall within the sensitivity range of forthcoming large-scale surveys, including DESI and *Euclid*. Observed deviations significantly larger or smaller than this level would directly challenge the proposed mechanism.

6.3. Primordial Saturation and Direct Collapse

The redshift dependence discussed above acquires particular significance in the primordial regime. As introduced in Section 2.4, the operational saturation scale

$$a_{\star}(z) \sim \kappa \frac{b_{\chi}}{\ell(z)} \quad (18)$$

depends inversely on the descriptive resolution $\ell(z)$. At very high redshift, the projected state remains spectrally constrained, so that $\ell(z)$ cannot decrease arbitrarily while preserving smooth quasi-injective updates.

Consider a collapsing primordial gas cloud of mass M and radius R . The characteristic update demand is set by the free-fall timescale,

$$D(R) \sim \frac{1}{t_{\text{ff}}} \sim \sqrt{\frac{GM}{R^3}}. \quad (19)$$

Fragmentation into stellar units requires that the characteristic fragmentation scale λ_{frag} remain larger than the minimal projectable scale $\ell(z)$. When $\lambda_{\text{frag}} < \ell(z)$, the projection cannot resolve a multi-body stellar configuration.

In this regime, admissibility can only be restored through a saturated reconfiguration channel. Direct collapse into a compact seed then becomes the structurally available outcome. This yields an implicit redshift-dependent critical scale $M_{\text{crit}}(z)$ defined by the condition that the required fragmentation scale equals $\ell(z)$. Massive halos above this threshold naturally enter a direct-collapse regime without requiring additional exotic physics.

6.4. Scaling Analysis of the Projective Gap

The inevitability of saturation can be established through a simple scaling argument. Let the update demand scale as

$$D(R) \propto R^{-3/2}, \quad (20)$$

as dictated by the free-fall timescale.

The projective capacity is limited by the maximal admissible throughput under the invariant bound b_{χ} . Denote by $C(R)$ the total admissible update capacity within a region of size R . Without assuming a specific geometric postulate, we parametrize

$$C(R) \propto R^n, \quad (21)$$

where $n \geq 0$ encodes the effective growth of independent relational channels permitted by the mapping Π at the epoch considered.

The projective stress ratio behaves as

$$\Xi(R) = \frac{D(R)}{C(R)} \propto R^{-(3/2+n)}. \quad (22)$$

For any physically reasonable $n \geq 0$, the ratio diverges as $R \rightarrow 0$. This guarantees the existence of a finite contraction scale R_{sat} at which $\Xi \sim 1$. Beyond this point, the demanded update rate exceeds the admissible projective throughput.

The result is robust with respect to the precise value of n . Even if capacity scaled volumetrically ($n = 3$), the divergence would remain unavoidable. Saturation during sufficiently deep collapse is therefore a structural feature of the framework rather than a model-dependent artifact.

6.5. Little Red Dots as Early Saturation Signatures

The bounded-response structure discussed throughout this work is not restricted to late-time, low-density environments. It also imposes consistency constraints in high-curvature primordial regimes.

Recent JWST observations have revealed a population of compact, luminous, high-redshift sources often referred to as “Little Red Dots” (LRDs) [18,19]. These objects have been interpreted as rapidly growing massive black holes, possibly embedded in dense ionized cocoons, suggesting compact configurations whose inferred masses and sizes challenge conventional early-growth timescales under standard modeling assumptions at redshifts $z \gtrsim 6$.

Within the present framework, such observations do not require modifications of early-universe physics. They are structurally compatible with the same bounded-response principle if primordial systems approach the maximal admissible projective capacity under extreme compression.

Two non-exclusive realizations are possible. First, halos exceeding a critical mass $M_{\text{crit}}(z)$ may undergo genuine direct collapse, producing massive compact seeds at early epochs. Second, a strongly constrained configuration may alter the effective geometric inference of mass profiles in high-curvature regimes, leading to enhanced apparent compactness without requiring proportionally large baryonic mass.

In both cases, the mechanism is identical at the effective level: a resolution gap between the demanded fragmentation scale and the minimal projectable scale $\ell(z)$. The observational signature is therefore not necessarily the presence of an exotic component, but the structural response of the inference metric near its saturation threshold.

This interpretation is introduced as a consistency check rather than a standalone prediction. The framework does not claim that LRDs uniquely validate the mechanism, but that their inferred mass estimates remain structurally compatible with the same bounded-response mechanism that governs the low-density Hubble inference discussed above.

Discriminating between a physical direct-collapse black hole and a predominantly projective overmass configuration requires combined constraints from variability, spectral diagnostics, and dynamical mass estimates. The framework predicts that the abundance of such compact early objects should track the redshift evolution of the effective resolution scale rather than solely the mass accretion history of halos.

We emphasize that this proposal does not assert incompatibility with standard early-universe cosmology. Rather, it provides a curvature-dependent consistency test linking primordial compact objects and late-time low-acceleration phenomena within a unified effective framework.

6.6. Void-Specific Redshift Drift

The effective saturation mechanism also affects the temporal evolution of redshift for sources embedded in low-density environments.

In the standard cosmological framework, the redshift drift depends solely on the global expansion history. In the present framework, sources located in deep voids experience

a modified effective transition rate, leading to a small but systematic deviation in the expected redshift drift signal.

Future high-precision spectroscopic experiments may be able to detect this effect by comparing sources located in voids with those in denser regions, complementing existing redshift-drift proposals [15]. The effect is predicted to be suppressed at high redshift, where environmental averaging dominates.

6.7. Weak Lensing Signatures in Voids

Because the effective framework modifies the inferred gravitational response in low-density regions, it also impacts weak gravitational lensing by cosmic voids.

The model predicts subtle, environment-dependent modifications in the inferred lensing convergence profiles of deep voids. We emphasize that no global inconsistency with weak lensing surveys is implied. In standard-density environments, where $g_N \gtrsim a_*$, the framework reduces to the unsaturated regime and reproduces conventional expectations. Potential deviations are restricted to the extreme low-acceleration outskirts of large voids, where the effective response approaches its saturation limit.

In this regime, the effect would manifest not as a breakdown of lensing consistency, but as a mild shift in the outer convergence profile relative to purely kinematic mass inferences. Such deviations remain within current observational uncertainties, but provide a discriminant target for future high-precision surveys probing the deepest underdense regions [15].

Void lensing measurements therefore provide an independent probe of the framework, complementary to kinematic tests, and have been identified as sensitive diagnostics of gravitational behavior in underdense regions [15].

6.8. Predictions for Ultra-Diffuse Galaxies

Ultra-diffuse galaxies constitute extreme low-surface-density systems and probe the saturated regime across most of their spatial extent [20].

Quantitatively, in the saturated regime ($g_N \ll a_*$), the effective framework predicts that the asymptotic circular velocity is bounded by the finite response of the effective gravitational field. To leading order, this implies

$$v_\infty \sim \sqrt{R a_*} [1 - \mathcal{O}(\delta_\rho)], \quad (23)$$

where R is the characteristic baryonic scale length of the system and $\delta_\rho = 1 - \rho_{\text{void}}/\rho_{\text{global}}$ is the local underdensity contrast. For ultra-diffuse galaxies in void environments ($\delta_\rho \approx 0.3$ – 0.6 and $R \sim 3$ – 5 kpc), this yields typical asymptotic velocities $v_\infty \lesssim 10$ – 15 km s^{−1}.

This prediction differs qualitatively from MOND, which enforces a universal relation $v_\infty \propto (M_{\text{baryon}} a_0)^{1/4}$ independent of environment. In the present framework, the asymptotic velocity depends explicitly on the large-scale environment through δ_ρ , leading to systematically lower velocities for ultra-diffuse galaxies in voids than for systems of comparable baryonic mass in denser regions.

Observations reveal a wide diversity of inferred dynamical mass content among ultra-diffuse galaxies, ranging from apparently dark-matter-rich systems to galaxies consistent with little or no dark matter [21,22]. Within the present framework, this diversity arises naturally from differences in how deeply individual systems probe the saturated regime.

In ultra-diffuse galaxies, the apparent mass discrepancy inferred from kinematics should therefore not be interpreted as evidence for an extended distribution of unseen matter. It reflects a geometric threshold effect. The baryonic configuration probes a regime in which further density dilution no longer increases the effective dynamical response.

As a result, the inferred dynamical mass tracks the onset of saturation rather than the presence of an additional gravitating component. The apparent “missing mass” is thus an emergent consequence of limited descriptive resolution in low-density systems.

This interpretation leads to a falsifiable prediction. If the inferred mass discrepancy originates from geometric saturation rather than from a material mass component, independent mass tracers should not reveal correspondingly extended matter distributions.

In particular, weak lensing signals, satellite dynamics, and stellar velocity dispersion profiles are expected to systematically underpredict the dynamical mass inferred from kinematic measurements in the deepest saturation regime. A positive detection of massive, spatially extended halos in ultra-diffuse galaxies would directly falsify the present framework.

Because ultra-diffuse galaxies probe the deepest saturation regime, they provide a particularly sensitive test of the universality of the saturation scale a_* .

6.9. Summary

The effective saturation framework yields a coherent and tightly constrained set of predictions across both galactic and cosmological scales.

All predictions arise from environmental dependence and involve no additional free parameters beyond the saturation scale a_* already fixed by rotation curve data.

The framework is therefore falsifiable with current or near-future observational capabilities. Confirmation or rejection of these signatures will decisively determine whether the proposed mechanism captures a genuine aspect of low-density gravitational phenomenology.

7. Conclusion

We have examined whether two long-standing observational anomalies, namely flat galaxy rotation curves and the Hubble constant tension, may originate from a common effective mechanism operating in low-density environments.

Within a minimal phenomenological framework characterized by a single saturation scale a_* , we have shown that baryonic matter alone reproduces the main features of observed galaxy rotation curves across a wide range of systems, without halo-by-halo tuning and without introducing dark matter particles at the level of the effective description. This result is supported by global fits to the SPARC sample.

The same saturation principle extends naturally to cosmological low-density regions. Cosmic voids probe the deepest saturation regime and provide a direct link between galactic low-acceleration phenomenology and late-time expansion inference. In such environments, the locally inferred expansion rate is predicted to deviate systematically from the global value inferred from early-universe observables, while the background Friedmann expansion remains unmodified.

Crucially, the framework does not alter early-time physics and does not modify the global cosmological dynamics. Gravitational acceleration and the Hubble parameter are interpreted as effective inference rates emerging from a bounded-response structure, rather than as fundamental fields requiring modification.

A distinctive strength of the proposal lies in its cross-sector consistency. The same empirically determined scale a_* governs both galactic rotation curves and environment-dependent expansion inference. Constraints obtained in one domain therefore directly restrict the allowed phenomenology in the other, preventing independent tuning of galactic and cosmological sectors.

The framework is sharply falsifiable. It predicts measurable correlations between inferred values of H_0 and large-scale density contrast, a redshift-dependent suppression of

the local offset, and specific signatures in ultra-diffuse galaxy kinematics, redshift drift, and void weak lensing. Failure of these correlated predictions would invalidate the mechanism.

The central claim of this work is therefore not that gravity is modified, nor that dark matter is replaced by a new fundamental constant, but that a bounded-response principle may underlie both galactic low-acceleration behavior and late-time expansion inference. Whether this effective description captures a genuine structural aspect of gravitational phenomenology or merely approximates deeper dynamics will be decided by forthcoming observational tests.

Appendix A Cosmological Operational Closure and the Scale $\ell = c/H$

This appendix derives Eq. (6) used in Section 2.5. The derivation relies solely on the operational closure axiom and the homogeneous saturated kinematics of the projected state. No geometric horizon concept is invoked.

Appendix A.1 A.1 Operational closure and descriptive resolution

Definition A.1 (Operational closure and descriptive resolution). The descriptive resolution ℓ is the minimal coarse-graining neighbourhood such that the update of the projected state U at epoch t depends on all substrate channels that can influence the update within a single admissible step, and on no others.

By definition A.1, the descriptive resolution ℓ is the minimal coarse-graining neighbourhood such that the update of the projected state U at epoch t depends on all substrate channels that can influence the update within a single admissible step, and on no others.

Closure therefore requires two properties:

1. Completeness: all admissible influences must be included.
2. Minimality: no inadmissible or causally irrelevant regions are included.

Appendix A.2 Homogeneous saturated kinematics

In the homogeneous late-time regime,

$$D_{\text{loc}}\chi_{\text{eff}} = c, \quad (\text{A24})$$

so propagation of relational influence is bounded by c .

The projected state evolves according to

$$H(t) = \frac{\dot{\chi}}{\chi}. \quad (\text{A25})$$

In a homogeneous regime, $H(t)$ is the unique intrinsic dimensionless rate. The characteristic update time is therefore

$$\tau(t) \sim \left(\frac{\dot{\chi}}{\chi}\right)^{-1} = H(t)^{-1}. \quad (\text{A26})$$

This is not imposed externally. It follows from the requirement that an update corresponds to an order-one change of the dimensionless global state variable.

Appendix A.3 Influence domain and lower bound

Within one update step of duration $\tau(t)$, the maximal propagation distance of relational influence is

$$r_{\text{max}} = c\tau(t) = \frac{c}{H(t)}. \quad (\text{A27})$$

Any substrate configuration at distance $r \leq r_{\max}$ can influence the projected state within the update step. Excluding such configurations from the coarse-graining neighbourhood would violate completeness of closure.

Therefore,

$$\ell \geq \frac{c}{H(t)}. \quad (\text{A28})$$

Appendix A.4 Minimality and upper bound

Conversely, any configuration at distance $r > r_{\max}$ cannot influence the update within the admissible step.

Including such regions in the neighbourhood does not modify the update map of U , but enlarges the coarse-graining domain beyond the minimal operationally closed set.

Minimality of closure therefore requires

$$\ell \leq \frac{c}{H(t)}. \quad (\text{A29})$$

Appendix A.5 Uniqueness of the homogeneous resolution

Combining the bounds yields

$$\ell_{\text{cosmo}}(t) = \frac{c}{H(t)}. \quad (\text{A30})$$

No additional scale is available in the homogeneous regime. Any alternative choice $\ell = f(H)$ would either violate dimensional consistency or introduce an extraneous parameter, contradicting the ontological minimality of the framework.

Appendix A.6 Consequence for the saturation scale

Substituting into

$$a_{\star}(t) \sim \kappa \frac{b_{\chi}}{\ell(t)} \quad (\text{A31})$$

gives

$$a_{\star}(t) \sim \kappa b_{\chi} \frac{H(t)}{c}. \quad (\text{A32})$$

In projectable regimes $b_{\chi} \rightarrow c$, yielding

$$a_{\star}(t) \sim \kappa c H(t). \quad (\text{A33})$$

This establishes a structural link between the galactic saturation scale and the cosmological expansion rate. The normalization factor κ is fixed independently by the U(1) holonomy condition discussed in Section 2.6.

Data Availability Statement: The galaxy rotation-curve data used in this work are drawn from the publicly available SPARC database [1]. The numerical code used to generate the galaxy rotation-curve figures and associated diagnostics is openly available and archived with a persistent identifier. The version used in this work is available via Zenodo at <https://doi.org/10.5281/zenodo.18462369> [?].

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